

Lesson 5: KINGDOM PROTISTA

All protists are eukaryotic, meaning they have a well-defined nucleus and membrane-bound organelles.

Most, but not all, protists are single-celled. Other than these features, they have very little in common.

- Protists are a diverse kingdom, including all eukaryotic organisms that are neither animals, nor plants, nor fungi.
- For classification, the protists are divided into three groups: animal-like protists, plant-like protists, and fungi-like protists.

Although Ernst Haeckel set up the Kingdom **Protista** in 1866, this kingdom was not accepted by the scientific world until the 1960s.

Characteristics of Protists

- Protists are unicellular
- They are eukaryotes (true nucleus)
- Colonial,
- Relative simple multicellular organisms that only rarely exhibit specialized cell types, or organs such as leaves or stems.
- Protist cells are eukaryotic with a double-membrane enclosed nucleus, double-stranded DNA, and specialized organelles, such as chloroplasts and mitochondria, in the cytoplasm.
- Some protists, such as Paramecium swim actively with beating hairs called cilia, amoeba..pseudopodia, euglena...flagella. Still other protists, such as the foraminiferans, are covered by a shell and often remain stationary.
- Many protists are photosynthetic while others are heterotrophic.
- Mostly found in aquatic environments..

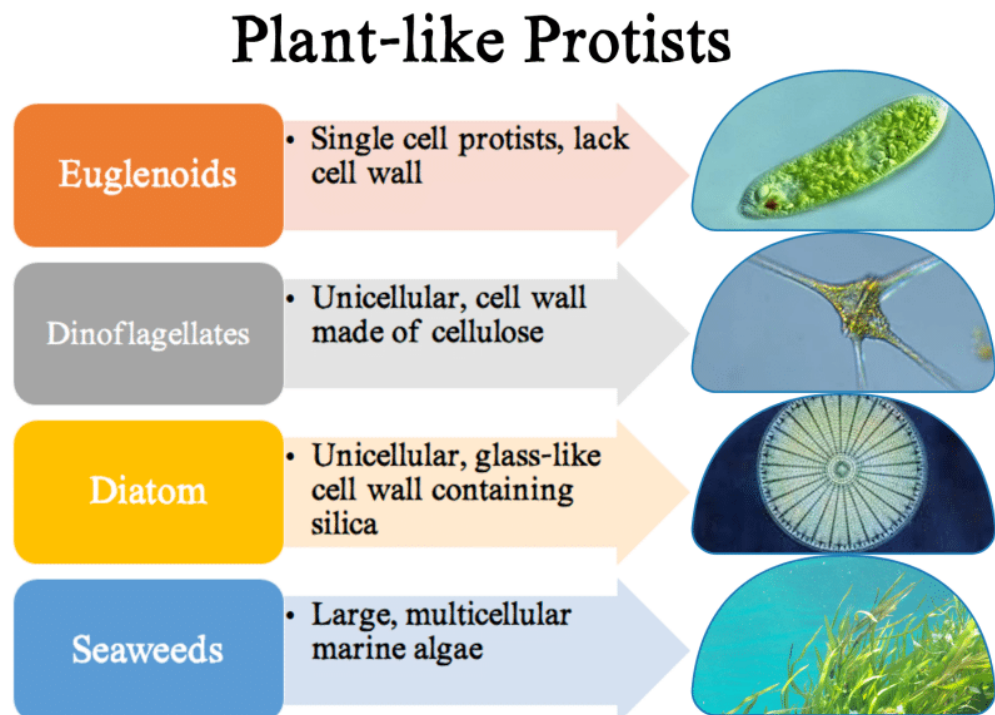
Types of Protists

Protists are broadly categorized into three informal groups based on how they obtain energy, which also reflects their biological characteristics:

➤ Plant-like Protists (Algae)

These protists contain chlorophyll and perform photosynthesis. They are essential primary producers in aquatic food webs. They can be unicellular, colonial, or multicellular, but they lack the complex tissues (like roots, stems, or leaves) found in true plants.

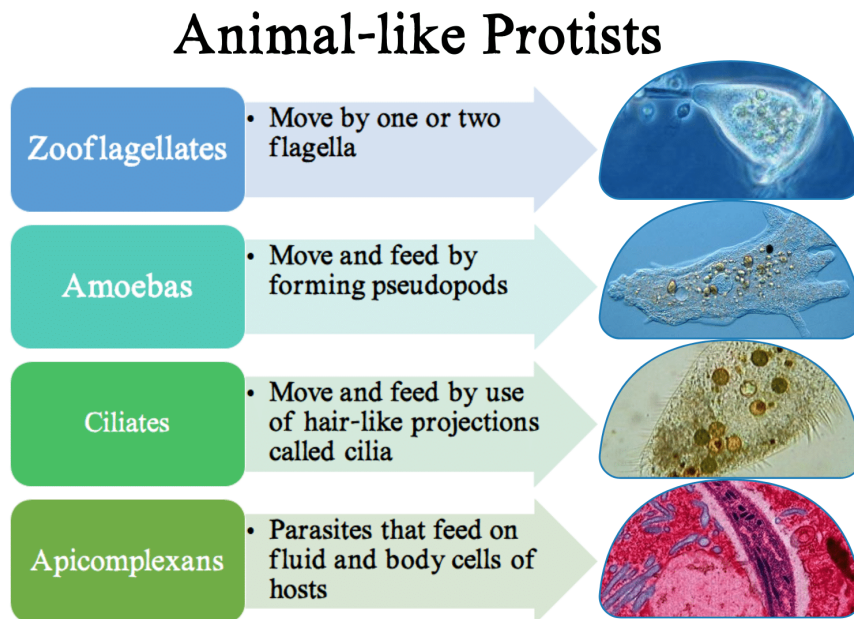
Examples: Diatoms, dinoflagellates, and giant kelp.



➤ **Animal-like Protists (Protozoa)**

These organisms are heterotrophic, meaning they must ingest food particles or absorb nutrients from their surroundings. They are typically unicellular and mobile, utilizing cilia, flagella, or pseudopodia to move and capture prey.

Examples: Amoebas, paramecia, and trypanosomes.



➤ **Fungus-like Protists**

These organisms often have a life cycle that resembles fungi; they absorb nutrients from dead or decaying organic matter and typically reproduce using spores. Unlike true fungi, which have cell walls made of chitin, fungus-like protists have cell walls made of cellulose (or sometimes no cell wall at all) and possess centrioles, which are absent in fungi. Slime molds are vibrant in color and are often found on rotting logs. Water molds are found in moist soil and on surface water. **Examples:** Slime molds and water molds.



PROTOZOANS

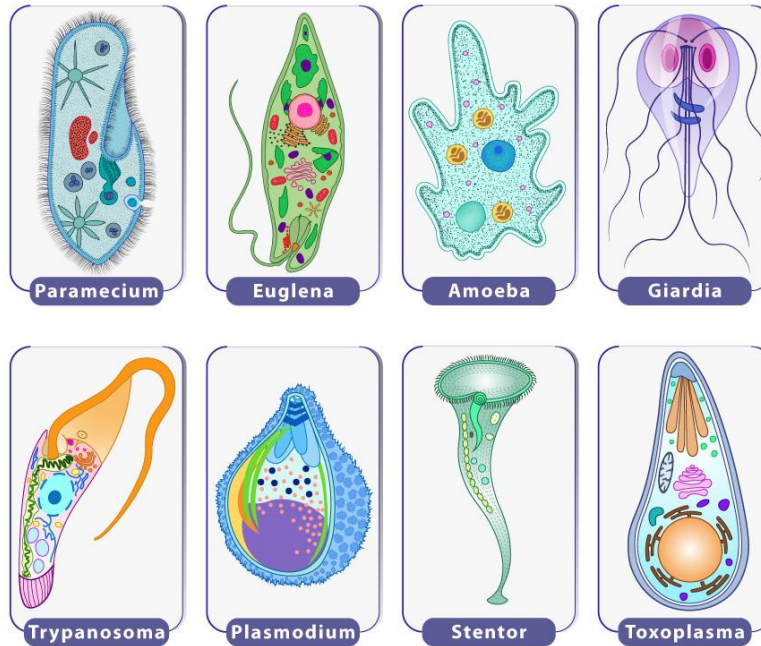
- The Protozoa are considered to be a subkingdom of the kingdom Protista, although in the classical system they were placed in the kingdom Animalia.
- More than 50,000 species have been described, most of which are free-living organisms; protozoa are found in almost every possible habitat.
- Protozoans are simple animals that play an important role in the ecosystem in which they are found.
- They can be described as motile, unicellular, wallless heterotrophic protists (*a protist is any eukaryotic organism that is not an animal, plant, or fungus*).
- They can be free-living predators or scavengers, ingesting other organisms or bits of organic matter, or parasitic or mutualistic symbionts. Protozoans simply mean.. earlier animals.
- Protozoa comprise of 8 phyla namely Rhizopoda, Foraminifera, Acrasiomycota, Myxomycota, Zoomastigina, Ciliophora, Apicomplexa and Oomycota.

- Most are microscopic. □
- All symmetries are present within members of the group. □
- No germ layers are present.
- No organs or tissues are formed, but specialized organelles serve many of these functions.
- They include free-living, mutualistic, commensal and parasitic forms.
- They move by pseudopodia, flagella, cilia and they can direct cell movements.
- All types of nutrition are present: autotrophic, heterotrophic and saprozoic(*parasitic ones These organisms do not have a "mouth" or the ability to engulf large particles. Instead, they absorb dissolved organic nutrients directly through their cell membrane via diffusion or active transport.*).
- They can be aquatic or terrestrial.
- Reproduction is asexual by fission, budding or cysts; or sexual by conjugation.

Subphylum / Class	Key Locomotory Structure	Primary Characteristics	Common Examples
Sarcomastigophora (Class Mastigophora)	Flagella	Move via whip-like tails; can be free-living or parasitic.	<i>Trypanosoma, Euglena</i>
Sarcomastigophora (Class Sarcodina)	Pseudopodia	Move by extending "false feet"; irregular, changing shape.	<i>Amoeba, Entamoeba</i>
Ciliophora	Cilia	Move via short, hair-like projections; possess dual nuclei (macro/micronucleus).	<i>Paramecium, Balantidium coli</i>
Apicomplexa (Sporozoa)	None (Adult stage)	Non-motile adults; obligate parasites; feature an "apical complex" for host penetration.	<i>Plasmodium</i>
Cnidospora / Microspora	None (Intracellular)	Highly specialized parasites that produce resistant spores with polar filaments.	<i>Nosema</i>

Protozoa

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Adaptations Of Protozoans To Their Environments

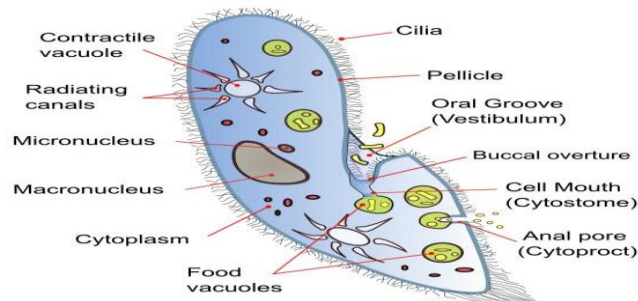
Protozoans have evolved a remarkable suite of adaptations that allow them to thrive in diverse environments, ranging from marine ecosystems and freshwater ponds to the internal tissues of host animals. Their survival strategies focus on three primary areas: locomotion, feeding/energy acquisition, and stress resistance.

➤ **Survival Mechanisms**

- **Encystment:** Many protozoans form resistant cysts when conditions are unfavorable (e.g., drought, cold, lack of food).
 - Cysts protect against desiccation, freezing, and nutrient scarcity.
 - They also aid in **dispersal**, as cysts can be carried by wind or water.
- **Excystment:** When conditions improve, protozoans break out of cysts and resume active life. Some species synchronize excystment with seasonal food availability.

➤ Structural Adaptations

- **Contractile Vacuoles:** Fresh-water protozoans live in a hypotonic environment, meaning water constantly tries to enter their cells by osmosis. They have evolved specialized **contractile vacuoles** that act as biological pumps, rhythmically expelling excess water to prevent the cell from bursting.
- **Pellicle:** A flexible yet supportive outer covering that maintains cell shape and provides protection. (A *pellicle* is a thin, flexible, or rigid proteinaceous layer beneath the plasma membrane in many protozoans).



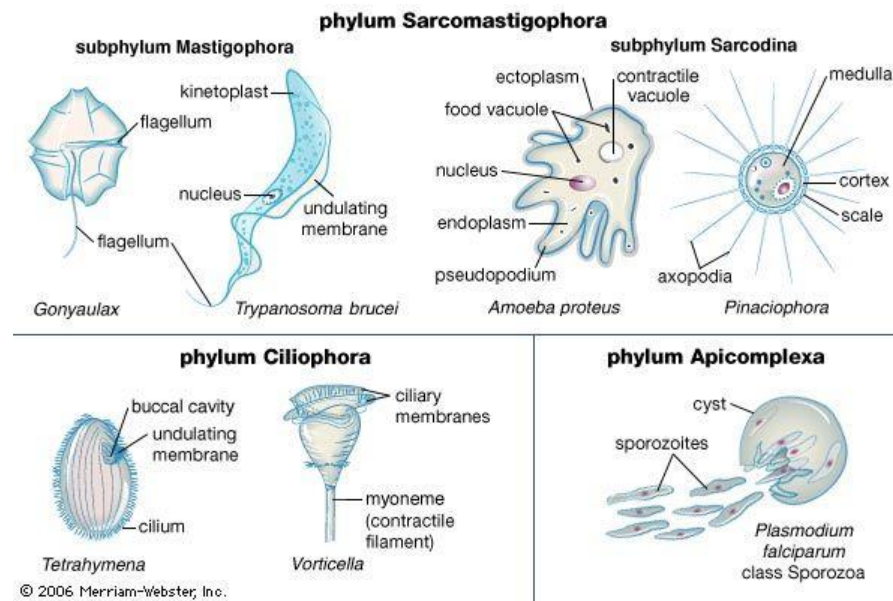
Specialized organelles:

- **Trichocysts/extrusomes** for defense or capturing prey. (Primarily used for **defense**, they are spindle-shaped vesicles found in the pellicle of *Paramecium* and other ciliates.)
- **Cytostome (cell mouth)** in ciliates for ingesting food.

➤ Locomotion Adaptations

Protozoans use different structures to move depending on their environment:

- **Pseudopodia:** Temporary extensions of cytoplasm used by amoebae for crawling and engulfing food.
- **Flagella:** Long whip-like appendages that propel organisms like *Euglena* and *Trypanosoma*.
- **Cilia:** Short, numerous hair-like structures that beat in coordination, allowing precise navigation in aquatic environments.

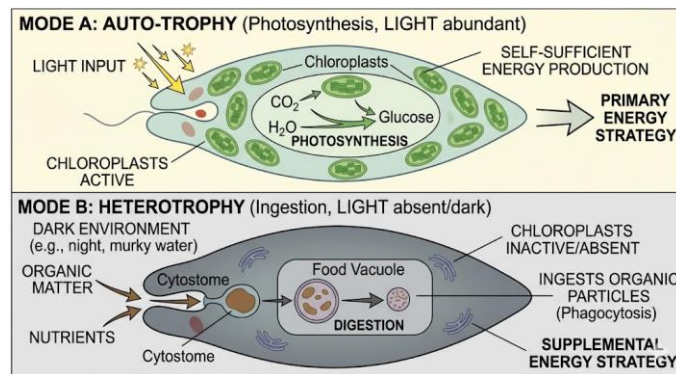


➤ Metabolic Adaptations

- **Mitochondria**: Provide energy for active lifestyles.
- **Modified mitochondria (mitosomes)**: Found in anaerobic protozoans like *Giardia*, adapted to oxygen-poor environments.
- **Symbiotic/parasitic lifestyles**: Some protozoans live inside hosts, benefiting from stable conditions and constant food supply.

➤ Nutritional Adaptations

- **Mixotrophy**: Many flagellates, such as *Euglena*, can switch between photosynthesis (when light is abundant) and heterotrophy (consuming organic matter in the dark). This is a critical adaptation for surviving in fluctuating environments where light or nutrient availability changes seasonally or daily.



What is the Importance of Protozoans?

Ecological Importance:

Protozoans form the vital link in the microbial food web, connecting primary producers (like algae) to larger organisms.

Energy Transfer: Protozoans act as primary consumers of bacteria and organic debris. By consuming these, they concentrate nutrients and biomass, which are then consumed by larger organisms like zooplankton, small fish, and invertebrates.

Nutrient Cycling: In both aquatic and soil environments, protozoans contribute to the decomposition process. They help release nitrogen and phosphorus back into the environment, which is crucial for the growth of plants and algae.

Water Quality Indicators: Because many protozoans are highly sensitive to environmental changes (pH, pollution, salinity), their presence—or absence—serves as a biological indicator of the health of a water body.

Medical and Pathogenic Importance

While many are beneficial, some protozoans are significant pathogens, and understanding them is crucial for global health.

Pathogen Identification: Studying parasitic protozoans like *Plasmodium* (malaria) or *Trypanosoma* (sleeping sickness) is fundamental to medical research. It has led to the development of anti-parasitic drugs and public health strategies to control the vectors (like mosquitoes) that transmit them.

Immunology Research: The study of how these parasites evade the host immune system (antigenic variation) has provided deep insights into how our own immune systems function and how pathogens evolve.

Scientific and Research Importance

Protozoans are the "workhorses" of biological research because they are unicellular, easy to culture, and perform all the life functions of a complex animal within one cell.

Model Organisms: Organisms like *Paramecium* and *Tetrahymena* have been used for decades to study fundamental biological processes, including cell division (mitosis/meiosis), genetics, and organelle function. Much of what we know about how eukaryotic cells operate was first discovered in protozoans.

Evolutionary Biology: Because they occupy so many different branches of the eukaryotic tree, studying their genomes helps scientists map the evolutionary history of all life on Earth.

Importance in Industry and Aquaculture

Protozoans also play a role in various economic sectors:

Waste Management: In sewage treatment plants, protozoans are intentionally used to feed on the bacterial population, helping to clarify the water and reduce the biological oxygen demand (BOD) before the water is returned to the environment.

Aquaculture Health: Understanding the protozoan population in fish ponds is essential for managing fish health. Monitoring for the presence of parasitic

species helps in early disease detection and prevents catastrophic stock losses in commercial farming.